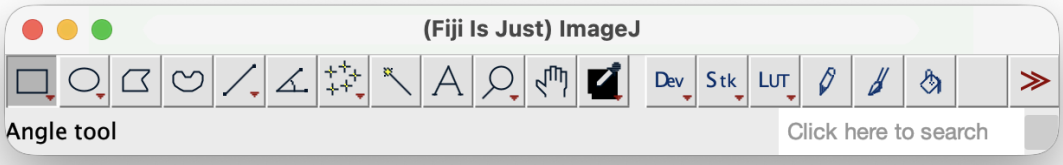
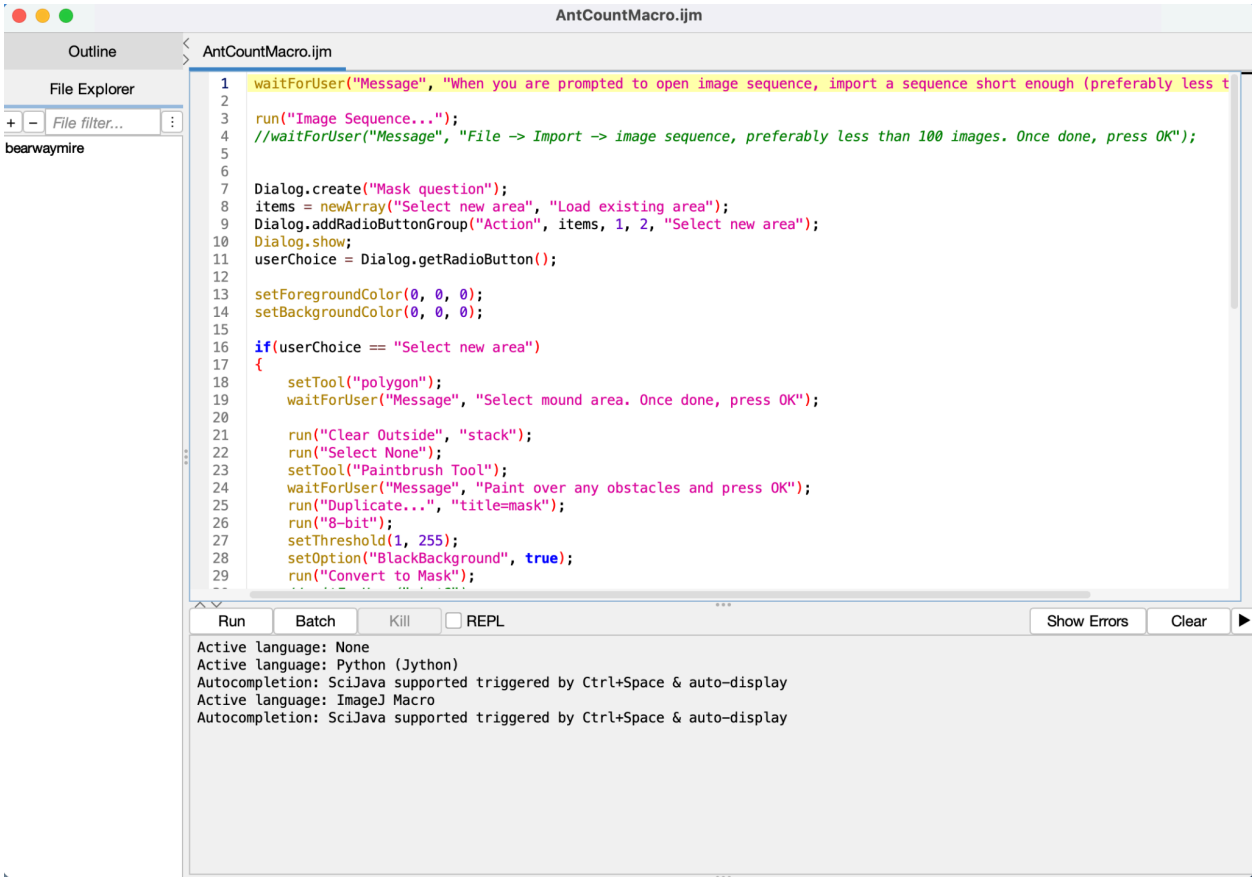
Overview: This guide provides step-by-step procedure for estimating ant activity on nest mounds using Fiji/ImageJ and Trackmate. There are two sections. First an ImageJ macro (ijm) is used to standardize image preparation (importing image sequences and creating masks), detect spots in a single window of images using TrackMate, and save the set parameters and number of detected spots as an XML file. Second, a Groovy script (.groovy) uses the previously saved XML file to apply the same detection parameters across multiple time windows in the same JPEG sequence.

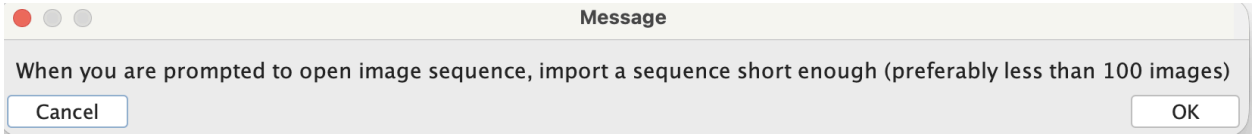
Section 1: Spot Detection in a Single Window and XML File Creation



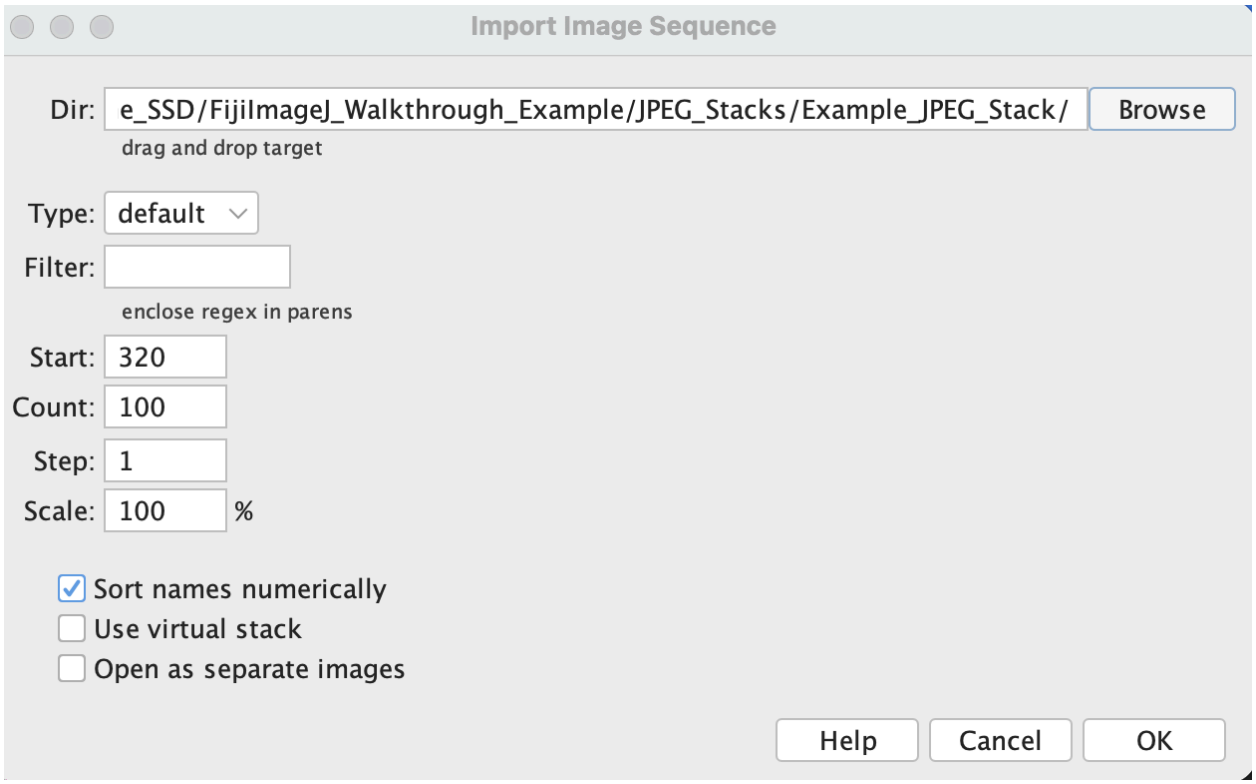
Step 1: Launch Fiji/ImageJ.



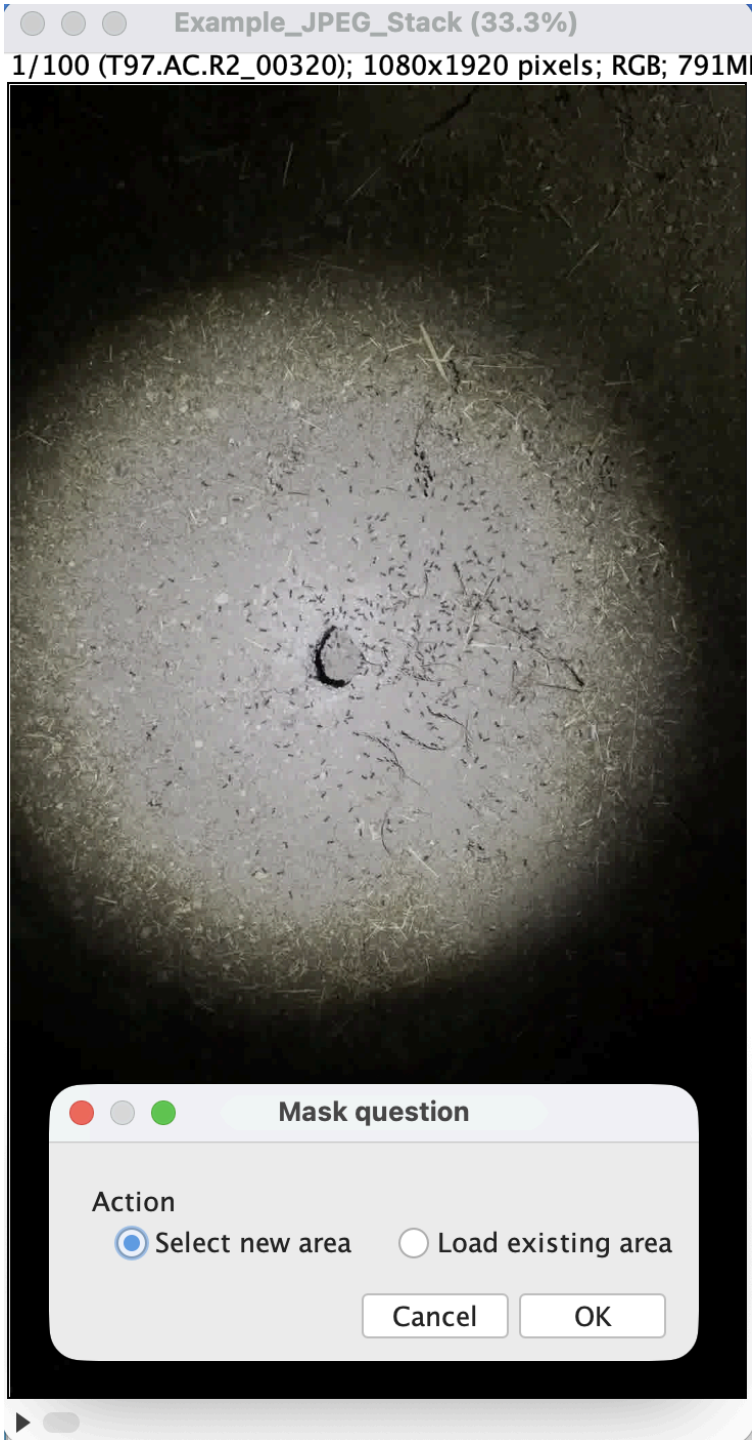
Step 2: Open the ImageJ macro (Initial_TrackMate_Analysis.ijm) in Fiji.



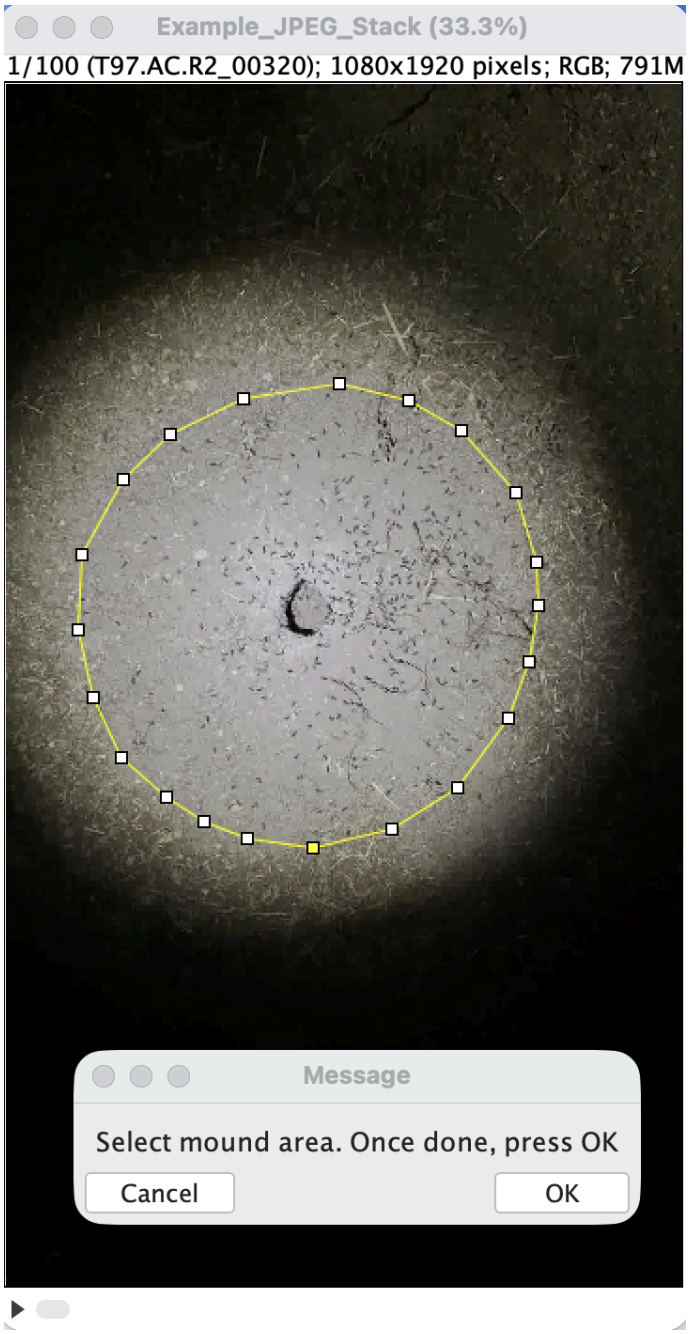
Step 3: Run the macro and follow the on-screen prompts.



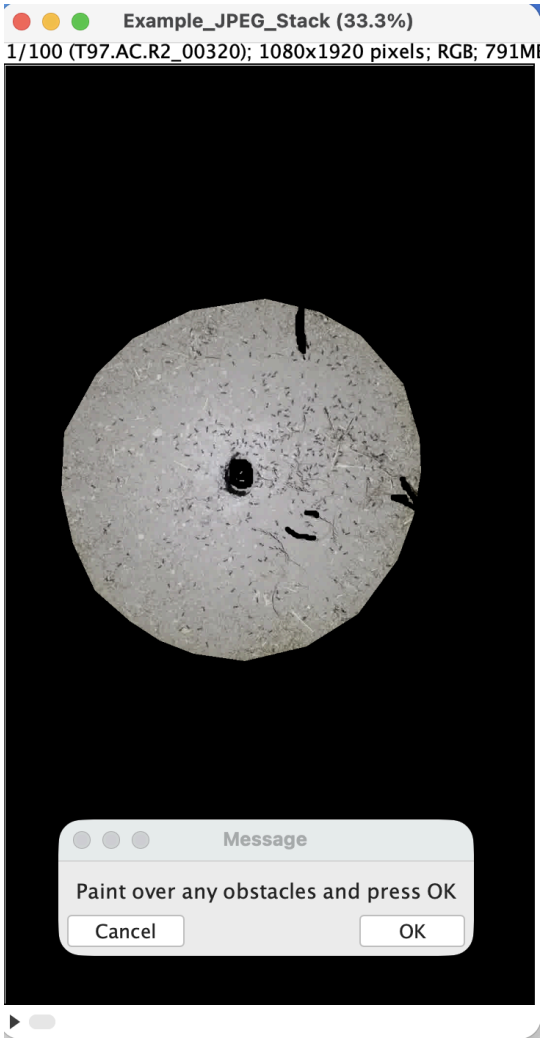
Step 4: Select the folder containing the JPEG image sequence and specify the starting frame and number of frames to process. For this project, videos were converted to JPEG sequences at 10 frames per second (fps). In this example, the starting frame is 320, corresponding to 32 seconds into the recording. Processing 100 frames corresponds to a ten second window from 32-42 seconds.



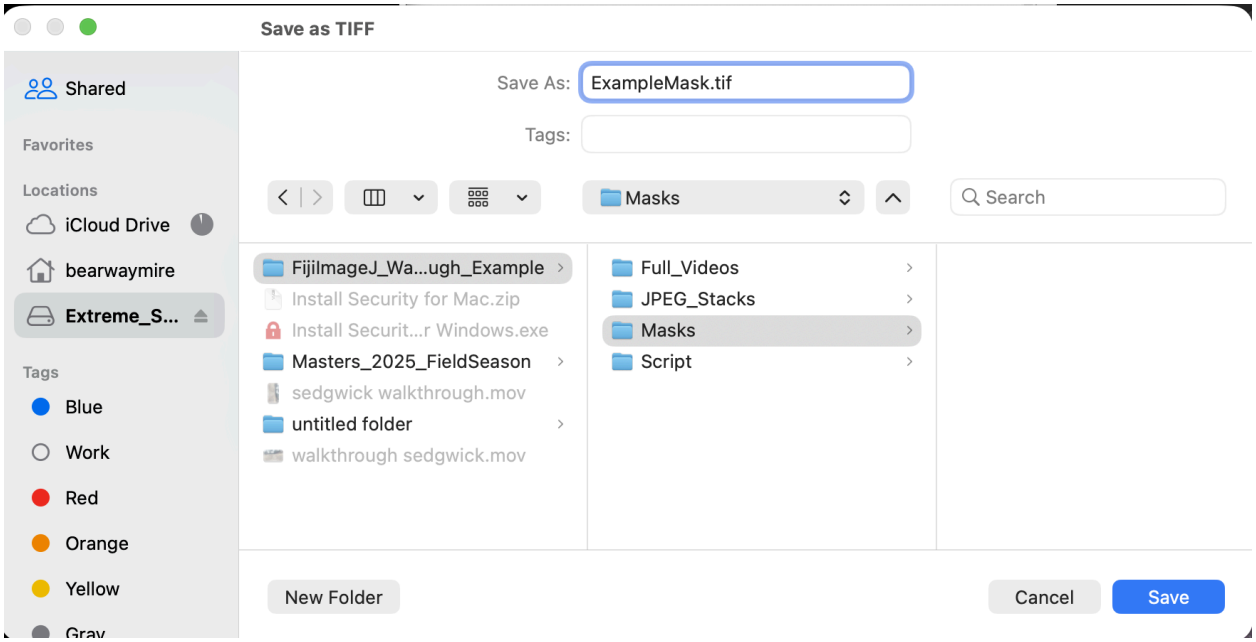
Step 5: Create a new mask defining the area of the mound to be analyzed, or load a previously used mask. Save the mask when prompted so you may reuse it across sampling windows from the same recording.



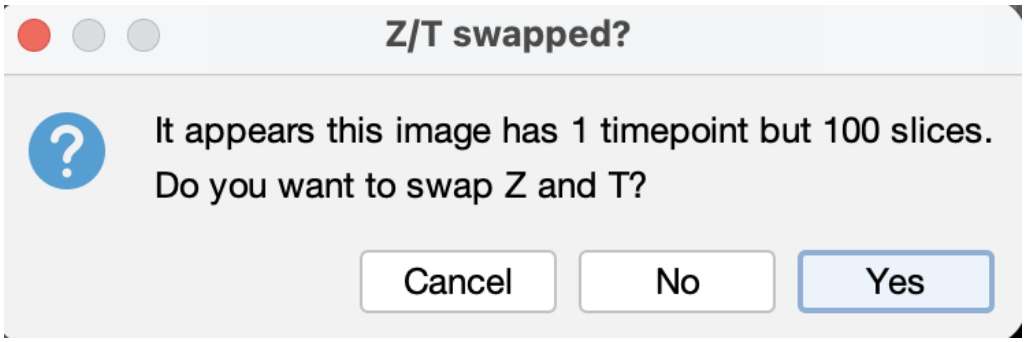
Step 6: Define the region of interest (ROI) by selecting points around the mound to create a polygon. Areas outside the selected polygon will be excluded from analysis.



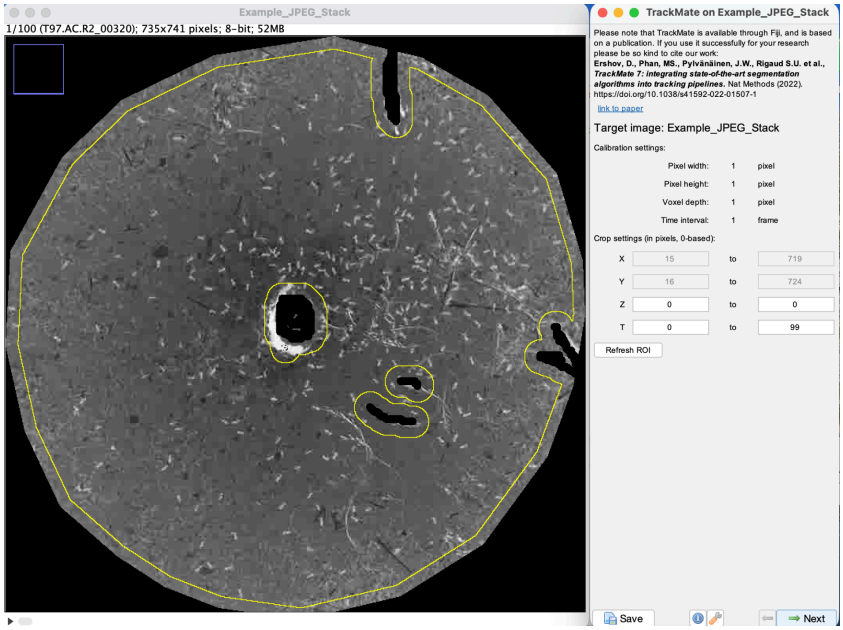
Step 7: Paint over any moving or bright obstacles (e.g., vegetation, bright surfaces, or large rocks) that may be incorrectly counted as spots by Trackmate. Once finished, clicking ok will prompt you to save the mask.



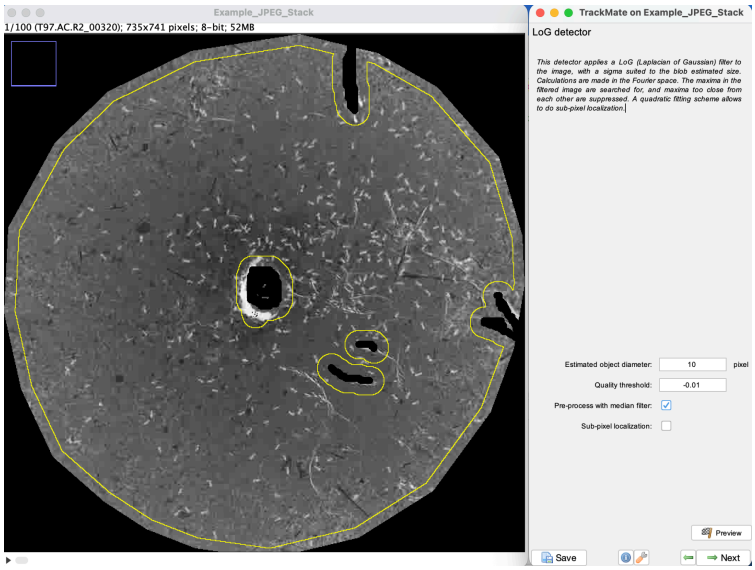
Step 8: Save mask



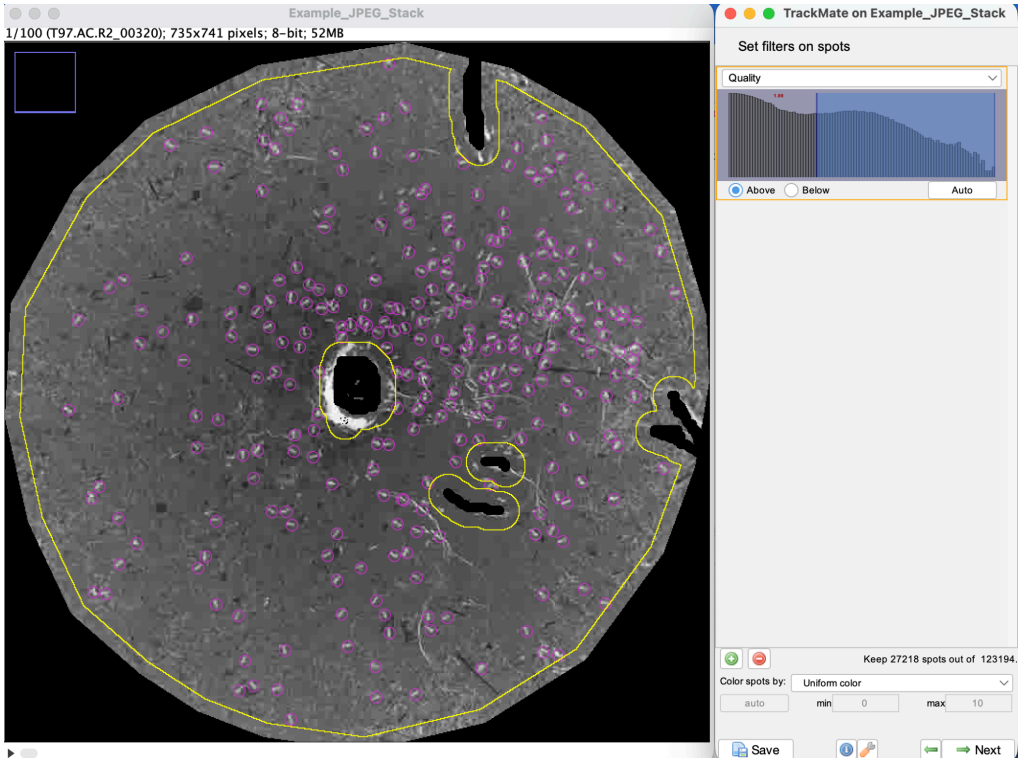
Step 9: Select ‘Yes’ to ensure TrackMate treats the imported images as consecutive frames of a video.



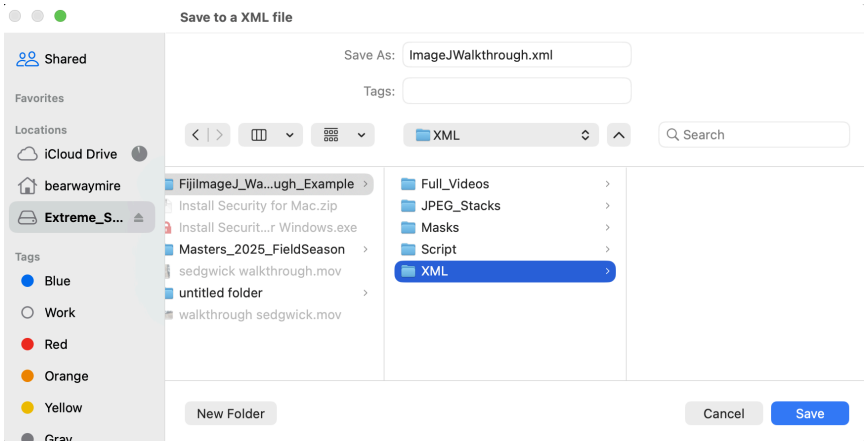
Step 10: Yellow boundary outlines will appear around your mask and painted regions, marking a 15-pixel buffer to help prevent unwanted spot detections along the edges of these areas.



Step 11: Click next to adjust the estimated object diameter according to the apparent size of the ants.

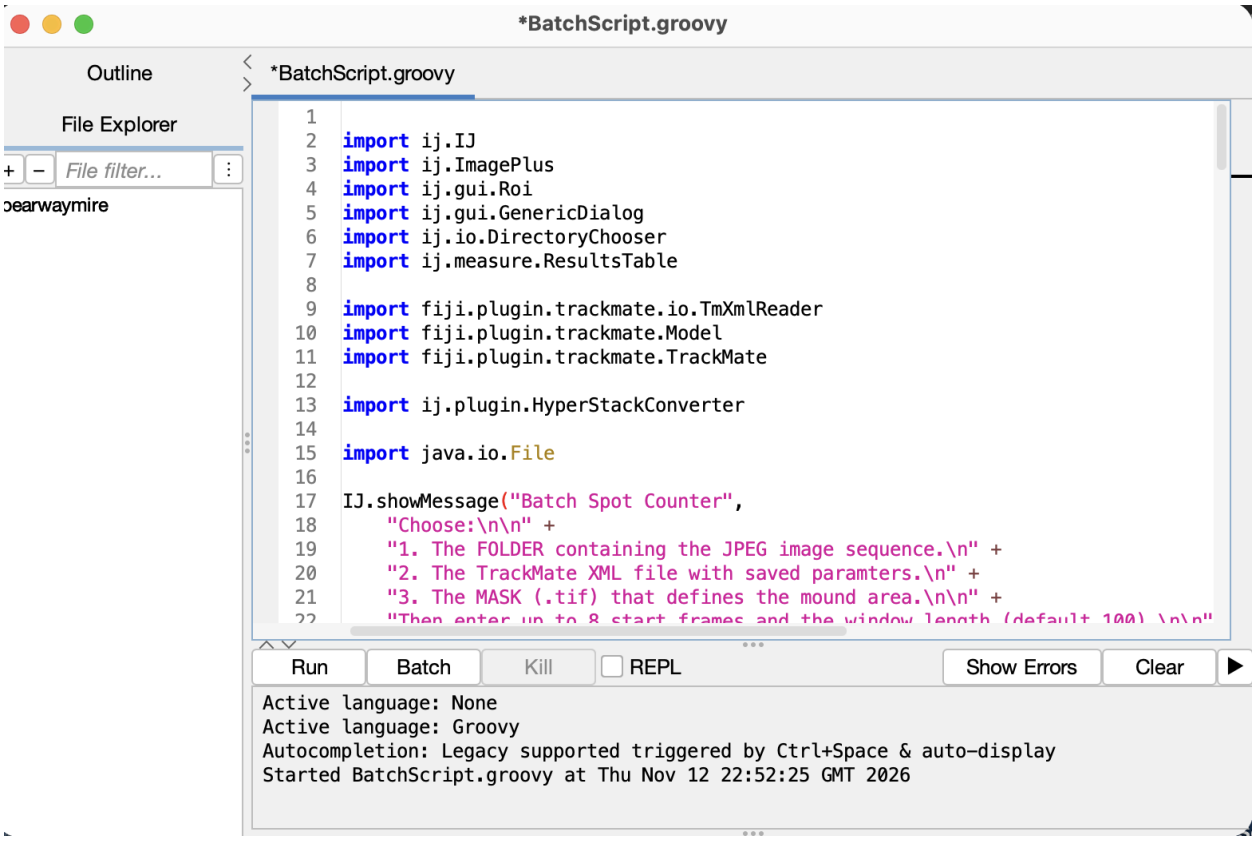


Step 12: Adjust the spot-quality threshold to retain spot detections that correspond to ants, while minimizing detections of other objects. The number on the bottom right (“keep x spots out of y”) indicates how many detected spots meet the selected quality threshold. In this example, 27218 spots were retained across 100 frames, corresponding to an average of 272.18 ants per frame over the 10-second sampling window.

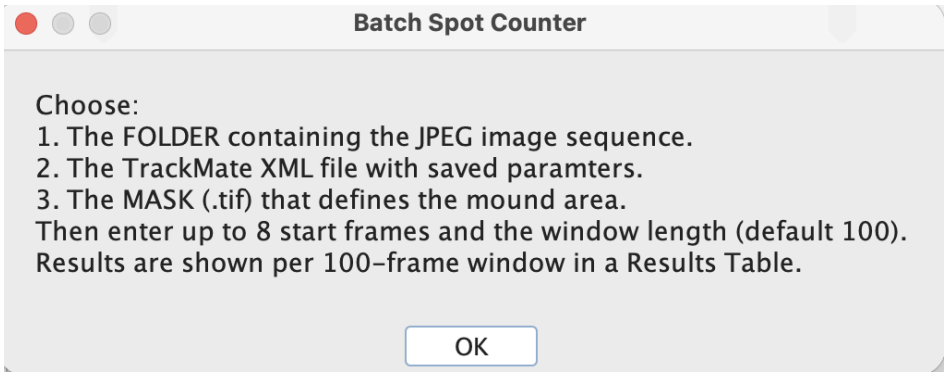


Step 13: Click ‘Save’ to save the analysis in an XML file. The file contains set parameters (including estimated object diameter and quality threshold) and detected spots. This XML file can be used to apply the same parameters to additional sampling windows.

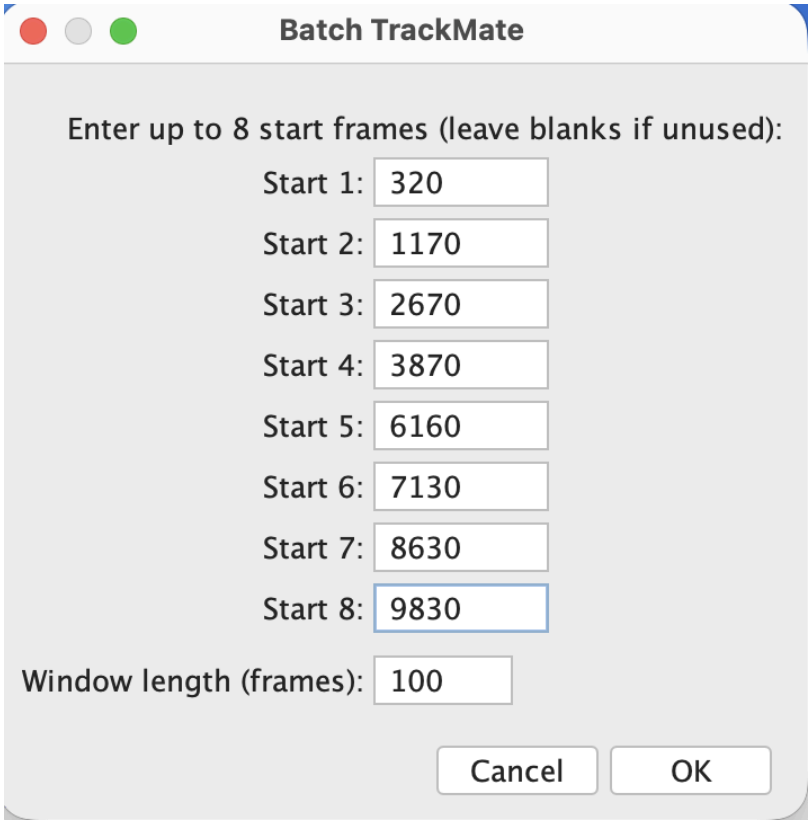
Section 2: Batch processing using saved XML file of TrackMate parameters



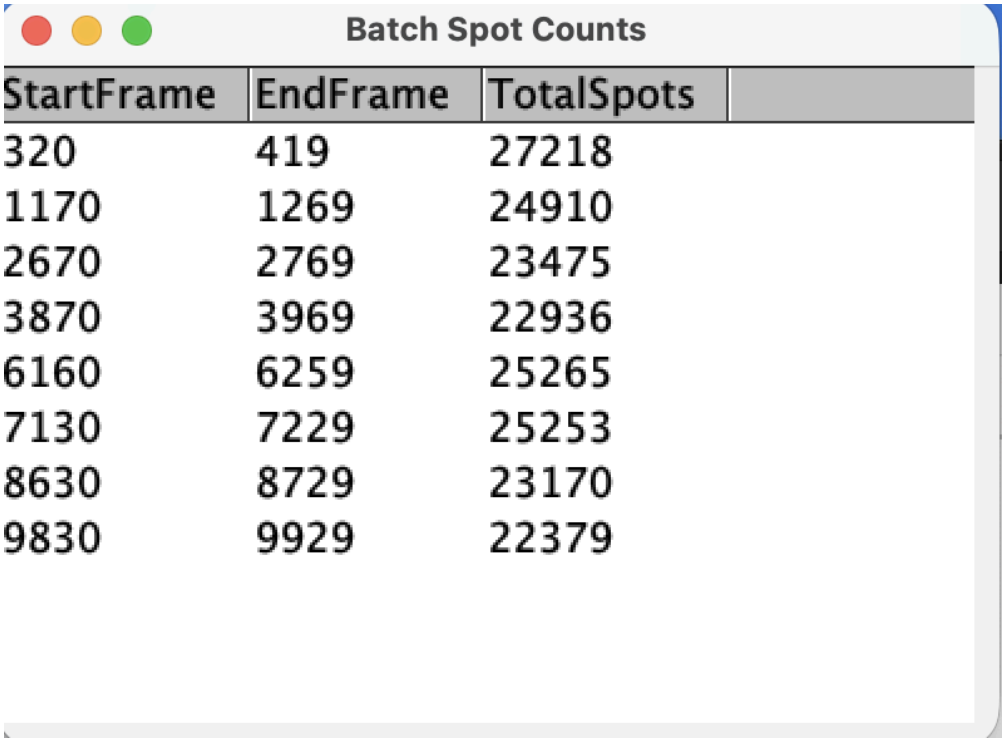
Step 14: Run the groovy script (Batch_Trackmate_Analysis.groovy) in Fiji's Script Editor to analyze additional sampling windows using the same parameters.



Step 15: When prompted, input this information, which will apply the same image-processing and detection settings to multiple sampling windows.



Step 16: Enter the starting frames of the sampling windows to be analyzed, as well as the number of frames per window. The script allows you to enter up to eight frames at once. Clicking OK will process all of them and produce an output window of your detected spots.



Step 17: Once processing is complete, a results table will display the total number of retained spots detected for each sampling window using the specified parameters and mask. Divide the total number of detections in each window frame by the number of frames (100 in our example) to calculate the mean number of detections per frame. If you include the same frame window used to create your XML file, you should get the same number of detected spots for that window.